

# SUMMER AI WORKSHOP

AI As Tutor, Study Partner,  
Explainer, Planner

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Business + Technology @





give me the answer

help me learn the thing



danger: passive fluency



**Answer Machine**  
definitions  
summaries  
solved examples  
polished plans



**Answer Machine** SKIPS

retrieving

reasoning

practicing

noticing confusion



## Learning Partner

asks what the student already knows

breaks the task into steps

gives hints before answers

checks understanding

adapts difficulty

asks the student to explain back



Do *not* ask AI to prove you are **done**

Ask AI to reveal what you still need to learn

# Prior Knowledge

Good tutoring starts by finding the learner's starting point

AI should ask: *What do you already know?*  
or give a short diagnostic quiz.

Without this, explanations are either too *easy*,  
too **hard**, or **falsely** reassuring



# Scaffolding

1. worked example
2. partially completed example
3. practice problem with hints
4. independent problem



# Socratic questions

What does this term mean?

Which variable changes?

Why would that conclusion follow?



# Retrieval practice

flash questions

mini-quizzes

"explain without notes" prompts

mistake-based review



Explica costo de oportunidad.



Sirve como tutor paciente para un estudiante universitario. Hazme tres preguntas sobre costo de oportunidad para diagnosticar mi nivel. Luego explica a mi nivel con un ejemplo cotidiano y uno de negocios. Luego, hazme prueba corta de tres preguntas. No me des las contestaciones hasta que lo intente. Si fallo, dame una pista antes de explicar completo.



## level

tutor paciente para un estudiante universitario

## goal

entender \*costo de oportunidad\*

## interaction

explicación con ejemplo cotidiano y de negocios

## feedback loop

tres preguntas para diagnosticar mi nivel y ajustar



"Explain this" -> content

"Tutor me through this" -> learning

Sirve de tutor para \_\_\_\_\_.

## ## Proceso

1. Preguntas para diagnosticar.
2. Explicar.
3. Prueba corta para verificar.
4. Corregir/retroalimentar.
5. Ayudar a crear plan para siguiente paso.



# Planning ingredients

Deadlines: exams, papers, quizzes, project milestones

Effort estimates: easy/medium/hard topics, realistic time blocks

Chunking: break “study Chapter 4” into smaller tasks

Self-tests: prove learning through recall and application

Study evidence: quiz score, solved problems, explain-back, draft

Adjustment: what changes if the student falls behind

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# Metacognition

thinking about *thinking*

I understood the explanation

I can solve a *new* problem without looking



# Integrity

Acceptable:  
asking for explanations  
quizzes  
feedback  
planning  
examples

*Risky:* asking AI to complete graded work.

**Dishonest:** submitting AI-produced thinking as your own.

What can I explain without notes?

What can I practice next?

What do I still not know?

My opinion: this is one of the highest-value lectures in the sequence. If students learn to use AI for diagnosis, retrieval, and feedback, they become better learners across every other class.

- DRIFT